

Rebecca Akeresola

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in linkedin

googlescholar

🌐 website

🏠 Edinburgh, UK

PROFESSIONAL PROFILE

- 📖 I am an emerging researcher with a PhD in Statistics, focusing on animal movement modelling for behavioural inference. I am currently a part of the Ecology and Marine Biology Research Group as a postdoctoral researcher at the University of Liverpool. My work lies at the interface of statistics and ecology, applying quantitative methods to analyse movement and environmental data in support of biodiversity conservation and evidence-based decision-making. My research interests include movement ecology, spatial statistics and the application of hidden Markov models to animal movement data.

RESEARCH EXPERIENCE

- 📖 **Postdoctoral Researcher** — *Dept. of Earth, Ocean and Ecological Sciences, University of Liverpool (2026 - present)*

Advised by: Prof. Samantha Patrick and Dr. Padget Oliver

Research focus: *Modelling animal movement for behavioural inference.*

- Research mainly on behavioral inference from Manx Shearwaters GPS-movement data using hidden Markov models (HMMs)
- Inferring homing behaviour from fine-scale movement data and understanding factors influencing homing behaviour

- 📖 **PhD Researcher** — *School of Mathematics, University of Edinburgh (2021-2025)*

Supervised by: Prof. Victor Elvira

Thesis title: *Modelling animal movement for behavioural inference.*

- Developed and validated hidden Markov models for animal movement analysis
- Applied models to seabird visual tracking data to infer behaviour and inform conservation
- Investigated the impact of telemetry data resolution on behavioural inference
- Produced outputs through first author publication and 15+ presentation talks
- Collaborative engagement across:
 - University of Edinburgh (UoE)
 - Biomathematics and Statistics Scotland (BioSS)
 - Joint Nature Conservation Committee (JNCC)
- Participated in an industry-led project (May 2021 - Jul 2021)

- 📖 **MSc Researcher** — *African Institute for Mathematical Sciences, Kigali, Rwanda (2019-2020)*

Supervised by: Prof. Ezra Gayawan

Dissertation title: *Bayesian Spatio-temporal Analysis of Greenhouse Gas Emissions (GHGs) in Africa.*

- Conducted Bayesian spatio-temporal modelling of greenhouse gas emissions across Africa
- Identified spatial and temporal trends relevant to policy and mitigation strategies

RESEARCH EXPERIENCE (continued)

- **BTech Researcher** — *Dept. of Statistics, Federal University of Technology Akure, Nigeria (2017)*
Supervised by: Prof. Ezra Gayawan
Dissertation title: *Analysis of the spatial patterns of malnutrition among women in Nigeria with a Bayesian structured additive model.*

- Conducted Bayesian spatial modelling of malnutrition among women in Nigeria
- Identified socio-economic determinants of regional health outcomes
- Produced outputs through first author publication

EDUCATION

- 2020 – 2025 ■ **PhD (with integrated study), Statistics**
University of Edinburgh and Heriot-Watt University, UK
- 2019 – 2020 ■ **MSc, Mathematical Sciences** (Distinction).
African Institute for Mathematical Sciences, Kigali, Rwanda
- 2012 – 2017 ■ **B.Tech, Statistics** (First Class Honours)
Federal University of Technology Akure, Nigeria

PEER-REVIEWED PUBLICATIONS

- 1 ■ **R. Akeresola**, A. Butler, E. Jones, R. King, V. Elvira, J. Black, and G. Robertson, “Validating hidden Markov models for seabird behavioural inference,” *Ecology and evolution*, vol. 14, 2024, [Citations: 5, Full-text views: 2140]. 🔗 DOI: 10.1002/ece3.11116
- 2 ■ **R. Akeresola** and E. Gayawan, “Analysis of the spatial patterns of malnutrition among women in Nigeria with a Bayesian Structured Additive Model,” *Geojournal*, vol. 85, pp. 81–92, 2020, [Citations: 16, Full-text views: 371]. 🔗 DOI: 10.1007/s10708-018-9958-0

PRESENTATIONS

- Delivered 15+ research presentations across seminars, conferences, and academic meetings:
Invited talks: 2 | Conference presentations : 2 | Seminar presentations: 11

Selected Engagements

- Young Statisticians’ Section of the Irish Statistical Association (Invited Talk, 2025)
- Joint Nature Conservation Committee (JNCC), Edinburgh (Invited Talk, 2023)
- National Centre for Statistical Ecology (NCSE) Meeting, Swansea (2023)
- Mathematical Biology Group Seminar, Edinburgh (2022, 2023, 2024)
- International Statistical Ecology Conference (ISEC), Swansea & South-Africa (2022, 2024)
- BioSS Annual General Meeting, Edinburgh (2022, 2023)
- MAC-MIGS CDT Symposium, Edinburgh (2021, 2022, 2023, 2024)

LEADERSHIP & ACADEMIC SERVICE

📌 **BioSS Representative — SEFARI Postgraduate Research Showcase, Edinburgh (2023)**

- Selected to represent Biomathematics and Statistics Scotland (BioSS) at The Scottish Environment Food and Agriculture Research Institute (SEFARI) Showcase
- Presented research to interdisciplinary audience

📌 **Session Chair — Centre for Statistics Annual Conference 2022, University of Edinburgh**

- Chaired part of an academic session ensuring structured delivery and time management
- Moderated discussion and facilitated Q&A

TEACHING & MENTORSHIP

📌 **Tutor — School of Mathematics, University of Edinburgh (2021–2024)**

- Delivered tutorials across 6 semesters to undergraduate and MSc students (10-20 per class)
- Provided student support and academic feedback for multiple courses

📌 **MSc Project Co-Supervisor — School of Mathematics, University of Edinburgh (Jul. – Aug. 2022)**

- Co-supervised projects in Statistics with Data Science course
- Guided research design, analysis, and interpretation

PROFESSIONAL DEVELOPMENT

📌 **Bayesian Filtering Summer School, Edinburgh, UK (2024)**

📌 **Continuous-Time Modelling Workshop, Edinburgh, UK (2023)**

📌 **Animal Movement Modelling Summer School, Sweden (2023)**

📌 **British Ecological Society Movement Ecology Conference (Attendance), St. Andrews, UK (2023)**

AWARDS, FUNDING & TRAINING

📌 **Funded Summer School on Animal Movement Modelling (2023)**

- Funded by Graduate School for Veterinary Medicine and Animal Sciences, Swedish University of Agricultural Sciences
- Selected participant for advanced training in animal movement modelling

📌 **EPSRC-funded PhD Studentship (2020–2024)**

Highly competitive funding awarded through MAC-MIGS Centre for Doctoral Training

📌 **African Institute for Mathematical Sciences (AIMS) Masters Scholarship (2019–2020)**

- Highly competitive fully funded scholarship awarded by AIMS-Rwanda
- Top candidates from across Africa chosen with < 6% selection rate.

PROFESSIONAL MEMBERSHIP

Member 📌 Royal Statistical Society | Membership No: 211473

RESEARCH & TECHNICAL SKILLS

Quantitative	Statistical modelling, Movement modelling, Spatio-temporal modelling
Data analysis	Large scale data handling, high-resolution movement data analysis, reproducible workflows (RMarkdown)
Programming	R (advanced), Python (intermediate), $\LaTeX$ (scientific writing and publishing)

REFEREES

Prof Samantha Patrick

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School of Environmental Sciences,
Dept. of Earth, Ocean and Ecological Sciences,
University of Liverpool,
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Dr Oliver Padget

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Prof Victor Elvira

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